

DATA STRUCTURE HW-1

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1. Sum of two matrices

Code

```
#include<stdio.h>
int main() {
    int r,c,i,j;
    printf("Enter rows and columns:");
    scanf("%d%d",&r,&c);
    int a[10][10],b[10][10],sum[10][10];
    printf("Enter elements of 1st matrix:\n");
    for(i=0;i<r;++i){
        for(j=0;j<c;++j){
            scanf("%d",&a[i][j]);
        }
    }
    printf("Enter elements of 2nd matrix:\n");
    for(i=0;i<r;++i){
        for(j=0;j<c;++j){
            scanf("%d",&b[i][j]);
        }
    }
    for(i=0;i<r;++i){
        for(j=0;j<c;++j){
            sum[i][j]=a[i][j]+b[i][j];
        }
    }
    printf("Sum of matrices:\n");
    for(i=0;i<r;++i){
        for(j=0;j<c;++j){
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

```
Enter rows and columns: 2
1
Enter elements of 1st matrix:
1
4
Enter elements of 2nd matrix:
6
4
Sum of matrices:
7
8
```

2. Read n number of values in an array & display them in reverse order

Code

```
#include<stdio.h>
int main(){
    int n,i;
    printf("Enter number of values:");
    scanf("%d",&n);
    int arr[100];
    printf("Enter %d values:",n);
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    printf("Reversed array:");
    for(i=n-1;i>=0;i--){
        printf("%d ",arr[i]);
    }
    return 0;
}
```

Output:

```
Enter number of values:3
Enter 3 values:4
3
21
Reversed array:21 3 4
```

3. Count the total number of duplicate elements in an array

Code

```
#include<stdio.h>
int main() {
    int n,i,j,count=0;
    printf("Enter size of array:");
    scanf("%d",&n);
    int arr[100];
    printf("Enter elements:");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                count++;
                break;
            }
        }
    }
    printf("Total duplicate elements:%d",count);
    return 0;
}
```

Output:

```
Enter size of array:3    4
Enter elements:12
34
56
23
Total duplicate elements:0
```

4. Print unique elements in an array

Code

```
#include<stdio.h>
int main() {
    int n,i,j;
    printf("Enter size of array:");
    scanf("%d",&n);
```

```

int arr[100];
printf("Enter elements:");
for(i=0;i<n;i++){
scanf("%d",&arr[i]);
}
printf("Unique elements:");
for(i=0;i<n;i++){
int unique=1;
for(j=0;j<n;j++){
if(i!=j&&arr[i]==arr[j]){
unique=0;
}
}
if(unique==1){
printf("%d ",arr[i]);
}
}
return 0;
}

```

Output:

```

Enter size of array:3
Enter elements:12
34
21
Unique elements:12 34 21

```

5. Insert in sorted array

Code

```

#include<stdio.h>
int main(){
int n,i,element;
printf("Enter number of elements:");
scanf("%d",&n);
int arr[100];
printf("Enter sorted elements:");
for(i=0;i<n;i++){
scanf("%d",&arr[i]);
}
printf("Enter element to insert:");
scanf("%d",&element);
for(i=n-1;i>=0&&arr[i]>element;i--){

```

```

arr[i+1]=arr[i];
}
arr[i+1]=element;
n++;
printf("Array after insertion:");
for(i=0;i<n;i++){
printf("%d ",arr[i]);
}
return 0;
}

```

Output:

```

Enter number of elements:4
Enter sorted elements:12
34
56

78
Enter element to insert:76
Array after insertion:12 34 56 76 78

```

6. Insert in unsorted array

Code

```

#include<stdio.h>
int main() {
int n,i,pos,element;
printf("Enter number of elements:");
scanf("%d",&n);
int arr[100];
printf("Enter elements:");
for(i=0;i<n;i++){
scanf("%d",&arr[i]);
}
printf("Enter position and element:");
scanf("%d%d",&pos,&element);
for(i=n;i>=pos;i--){
arr[i]=arr[i-1];
}
arr[pos]=element;
n++;
printf("Array after insertion:");
for(i=0;i<n;i++){
printf("%d ",arr[i]);
}

```

```
}  
return 0;  
}
```

Output:

```
Enter number of elements:3  
Enter elements:12  
43  
12 3  
Enter position and element:2  
34  
Array after insertion:12 3 43 13
```

7. Matrix multiplication

Code

```
#include<stdio.h>  
int main() {  
    int r1,c1,r2,c2,i,j,k;  
    printf("Enter rows and columns for first matrix:");  
    scanf("%d%d",&r1,&c1);  
    printf("Enter rows and columns for second matrix:");  
    scanf("%d%d",&r2,&c2);  
    if(c1!=r2){  
        printf("Multiplication not possible.\n");  
        return 0;  
    }  
    int a[10][10],b[10][10],mul[10][10];  
    printf("Enter elements of matrix 1:\n");  
    for(i=0;i<r1;i++){  
        for(j=0;j<c1;j++){  
            scanf("%d",&a[i][j]);  
        }  
    }  
    printf("Enter elements of matrix 2:\n");  
    for(i=0;i<r2;i++){  
        for(j=0;j<c2;j++){  
            scanf("%d",&b[i][j]);  
        }  
    }  
    for(i=0;i<r1;i++){  
        for(j=0;j<c2;j++){  
            mul[i][j]=0;  
            for(k=0;k<c1;k++){
```

```

mul[i][j]+=a[i][k]*b[k][j];
}
}
}
printf("Multiplied matrix:\n");
for(i=0;i<r1;i++){
for(j=0;j<c2;j++){
printf("%d ",mul[i][j]);
}
printf("\n");
}
return 0;
}

```

Output:

```

Enter rows and columns for first matrix:2
2
Enter rows and columns for second matrix:2
2
Enter elements of matrix 1:
23
34
56
67
Enter elements of matrix 2:
67
89
23  1
90
Multiplied matrix:
2255 5107
5159 11014

```

8. Matrix transpose

Code

```

#include<stdio.h>
int main(){
int r,c,i,j;
printf("Enter rows and columns:");
scanf("%d%d",&r,&c);
int a[10][10],t[10][10];
printf("Enter elements:\n");
for(i=0;i<r;i++){
for(j=0;j<c;j++){

```

```

scanf("%d",&a[i][j]);
}
}
for(i=0;i<r;i++){
for(j=0;j<c;j++){
t[j][i]=a[i][j];
}
}
printf("Transpose of matrix:\n");
for(i=0;i<c;i++){
for(j=0;j<r;j++){
printf("%d ",t[i][j]);
}
printf("\n");
}
return 0;
}

```

Output:

```

Enter rows and columns:2
2
Enter elements:
23
98
19
69
Transpose of matrix:
23 19
98 69

```

9. Right diagonal sum

Code

```

#include<stdio.h>
int main(){
int n,i,j,sum=0;
printf("Enter size of square matrix:");
scanf("%d",&n);
int a[10][10];
printf("Enter elements:\n");
for(i=0;i<n;i++){
for(j=0;j<n;j++){
scanf("%d",&a[i][j]);
}
}
for(i=0;i<n;i++){

```



```

for(j=0;j<n;j++){
if(i==j){
sum+=a[i][j];
}
}
}
printf("Right diagonal sum:%d\n",sum);
return 0;
}

```

Output:

Enter size of square matrix:2

Enter elements:

76

35

97

86

Right diagonal sum:162

10. Check for identity matrix

Code

```

#include<stdio.h>
int main() {
int n,i,j,flag=1;
printf("Enter size of square matrix:");
scanf("%d",&n);
int a[10][10];
printf("Enter elements:\n");
for(i=0;i<n;i++){
for(j=0;j<n;j++){
scanf("%d",&a[i][j]);
}
}
for(i=0;i<n;i++){
for(j=0;j<n;j++){
if(i==j&& a[i][j]!=1){
flag=0;
}
if(i!=j&& a[i][j]!=0){
flag=0;
}
}
}
}

```

```

}
if(flag==1){
printf("It is an identity matrix.\n");
}
else{
printf("It is not an identity matrix.\n");
}
return 0;
}

```

Output:

```

Enter size of square matrix:2
Enter elements:
2
3
5
7
It is not an identity matrix.

```

11. Find no. of vowels, consonants, digits and white space

Code

```

#include<stdio.h>
int main(){
char str[150];
int v=0,c=0,d=0,s=0,i;
printf("Enter string:");
fgets(str,sizeof(str),stdin);
for(i=0;str[i]!='\0';i++){
if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U'){
v++;
}
else if((str[i]>='a'&&str[i]<='z')||(str[i]>='A'&&str[i]<='Z')){
c++;
}
else if(str[i]>='0'&&str[i]<='9'){
d++;
}
else if(str[i]==' '){
s++;
}
}
printf("Vowels:%d Consonants:%d Digits:%d Spaces:%d",v,c,d,s);

```

```
return 0;
}
```

Output:

```
Enter string:ssak the great
Vowels:4 Consonants:8 Digits:0 Spaces:2
```

12. Frequency of a character in a string

Code

```
#include<stdio.h>
int main() {
    char str[100],ch;
    int count=0,i;
    printf("Enter a string:");
    fgets(str,sizeof(str),stdin);
    printf("Enter a character to find its frequency:");
    scanf("%c",&ch);
    for(i=0;str[i]!='\0';i++){
        if(ch==str[i]){
            count++;
        }
    }
    printf("Frequency of %c:%d",ch,count);
    return 0;
}
```

Output:

```
Enter a string:ssak
Enter a character to find its frequency:s
Frequency of s:2
```

13. Count words in a string

Code

```
#include<stdio.h>
int main() {
    char str[200];
    int i,words=1;
```

```

printf("Enter a string:");
fgets(str,sizeof(str),stdin);
for(i=0;str[i]!='\0';i++){
if(str[i]==' '&&str[i+1]!=' '){
words++;
}
}
printf("Number of words:%d",words);
return 0;
}

```

Output:

```

Enter a string:ssak
Number of words:1

```

14. Sort string array

Code

```

#include<stdio.h>
#include<string.h>
int main(){
char str[10][50],temp[50];
int n,i,j;
printf("Enter number of strings:");
scanf("%d",&n);
printf("Enter strings:\n");
for(i=0;i<n;i++){
scanf("%s",str[i]);
}
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(strcmp(str[i],str[j])>0){
strcpy(temp,str[i]);
strcpy(str[i],str[j]);
strcpy(str[j],temp);
}
}
}
printf("Sorted strings:\n");
for(i=0;i<n;i++){
printf("%s\n",str[i]);
}
return 0;
}

```

```
}
```

Output:

Enter number of strings:4

Enter strings:

Sri

Senthil

Adavan

K

Sorted strings:

Adavan

K

Senthil

Sri

15. Toggle case of a sentence (lowercase to UC & vice versa)

Code

```
#include<stdio.h>
int main(){
char str[100];
int i;
printf("Enter a sentence:");
fgets(str,sizeof(str),stdin);
for(i=0;str[i]!='\0';i++){
if(str[i]>='A'&&str[i]<='Z'){
str[i]=str[i]+32;
}
else if(str[i]>='a'&&str[i]<='z'){
str[i]=str[i]-32;
}
}
printf("Toggled sentence:%s",str);
return 0;
}
```

Output:

Enter a sentence:w What Sentence to WRItE

Toggled sentence:wHAT sENTENCE TO wriTe